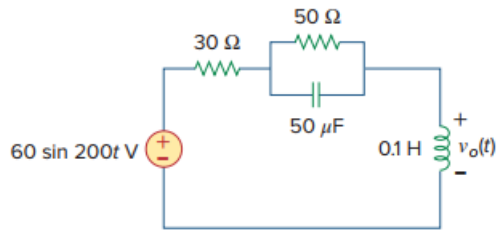


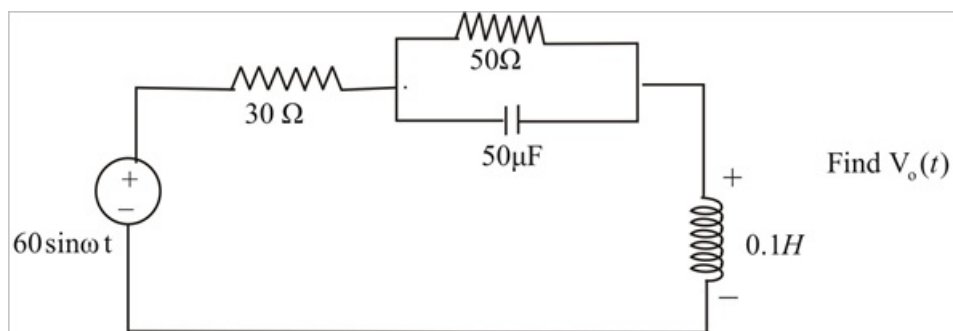
Problem

Calculate $v_o(t)$ in the circuit of Fig. 9.49.



Step-by-step solution

Step 1 of 4



Step 2 of 4

$$V = 60 \sin 200t = 60 \cos(200t - 90^\circ)$$

In polar form,

$$= 60 \angle -90^\circ$$

$$\omega = 200$$

$$Z_1 = 30$$

$$Z_2 = 50 \Omega \parallel 50 \mu F$$

$$Z_2 = \frac{50 \left(\frac{1}{j 200 \times 50 \times 10^{-6}} \right)}{50 + \frac{1}{j 200 \times 50 \times 10^{-6}}}$$

$$Z_2 = \frac{-j 5000(50 + j 100)}{12500}$$

$$Z_2 = 40 - j 20$$

Step 3 of 4

$$\text{And } Z_3 = j \times 200 \times 0.1 = j20$$

Current in the circuit,

$$V = I Z_{eq}$$

$$Z_{eq} = Z_1 + Z_2 + Z_3$$

$$= 30 + 40 - j20 + j20$$

$$= 70$$

$$I = \frac{60 \angle -90^\circ}{70}$$

$$I = 0.857 \angle -90^\circ$$

Step 4 of 4

Also,

$$\Rightarrow V_0 = I Z_3$$

$$= \{0.857 \angle -90^\circ\} \{20 \angle 90^\circ\}$$

$$= 17.14 \angle 0^\circ$$

In time domain,

$$V_0(t) = 17.14 \cos(200t) \text{ .}$$